Hierarchical Augmentation and Region-Aware Contrastive Learning for Semi-Supervised Semantic Segmentation of Remote Sensing Images

Yuan Luo, Bin Sun, *Member, IEEE*, Shutao Li, *Fellow, IEEE*, and Yulong Hu

***Abstract*— Semi-supervised semantic segmentation has gained significant attention as a method to reduce the substantial expense associated with pixel-level labeling. The existing methods primarily rely on consistency regularization or self-training. Recent consistency regularization methods augment the input images with weak or strong augmentation (SA) to improve the performance. However, such simple augmentations are not sufficient to simulate the variations in remote sensing images. The self-training methods exclude the noisy pseudo labels by some selection from the unsupervised training process to obtain better performance. The selection may lead to semantic information loss and bias of latent distribution. To solve the above two problems, we propose hierarchical augmentation (HA) and region-aware contrastive (RC) learning, namely HARC, for remote sensing images. The HA strategy simulates three levels of remote sensing image variations, i.e., spatial variations, uniform spectral variations, and uneven spectral variations. It can significantly enhance the model's capability to handle more intricate variations. The RC learning learns a class-wise feature distribution of all unlabeled samples instead of some screened unlabeled samples. It can eliminate semantic information loss and enhance the model's resistance to noise from pseudo labels. Our method is evaluated on three public remote sensing datasets, and the experimental results demonstrate its superiority over state-of-the-art (SOTA) semi-supervised methods.**

***Index Terms*— Hierarchical augmentation (HA), region-aware contrastive (RC) learning, remote sensing, semi-supervised semantic segmentation.**

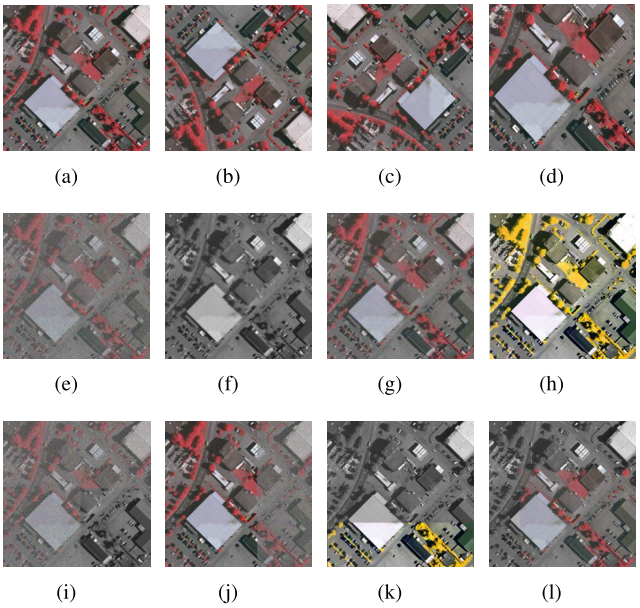


Fig. 1. Different augmentations corresponding to complex variations in remote sensing images. (a) Original image. (b)–(d) Scenarios with spatial variations. (e)–(h) Scenarios with uniform spectral variations. (i)–(l) Scenarios with uneven spectral variations.

I. Introduction

With the rise of deep learning [1], [2], many outstanding works [3], [4], [5], [6], have emerged in semantic segmentation. However, their success all has a common premise: the requirement for plenty of refined pixel-level labels for supervision. Producing these labels is so expensive that it hinders the application of these technologies in many scenarios, like cropland segmentation and urban planning. Therefore, semi-supervised semantic segmentation using only a limited quantity of labeled data and a substantial volume of unlabeled data has attracted widespread attention [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27]. Designing a reliable training strategy for unlabeled data is crucial in semi-supervised learning. Recent methods are mainly divided into two categories: consistency regularization [7], [8], [9], [10], [11], [12], [13], [14] and self-training [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27].

Consistency regularization enables two different augmented versions of the same image to produce consistent predictions. Recently, strong augmentation (SA) has been shown to be beneficial in enhancing the performance of models trained on unlabeled data [7]. Therefore, some methods based on weak-to-strong consistency regularization have been proposed [8], [9], [10], [11]. In weak-to-strong consistency regularization, the prediction for the weak augmentation (WA) samples is considered more accurate than the prediction for the SA

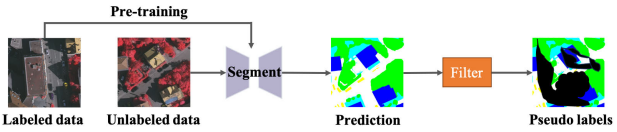
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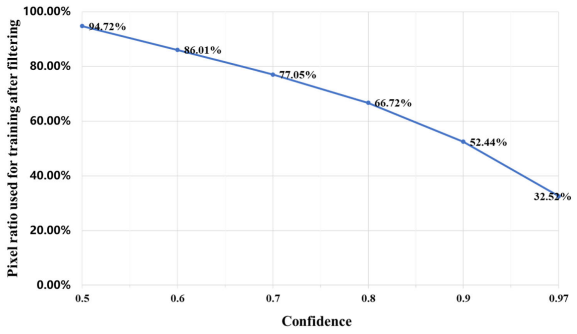
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(a)



(b)

Fig. 2. Self-training structure and the diagram illustrating the correlation of confidence with the number of pseudo-labels. (a) General paradigm of confidence selecting pseudo labels. (b) Proportion of available pseudo labels at different confidence. Note, the network is an FNC with resnet-50 trained on 1/4 labeled Vaihingen dataset. The proportions are collected on the 3/4 unlabeled Vaihingen dataset.

samples. So, using the prediction for the WA samples as the pseudo label of the SA samples could allow the model to be well trained. In these methods, WA consists of some geometric augmentation strategies, like rotate, resize, and flip. It can simulate the imaging influences caused by spatial variations. As shown in Fig. 1(b)–(d). SA consists of some iconographic augmentation strategies, like blur, noise, and gray. It can simulate the imaging influences caused by spectral variations. As shown in Fig. 1(e)–(h). It is noticeable that such simple augmentations are not sufficient to simulate the influence on remote sensing images, which also have some uneven and more complex spectral variations, like Fig. 1(i)–(l). Therefore, we believe that adding a path to simulate uneven spectral variations based on the existing weak-to-strong consistency regularization may be helpful to improve model performance.

Self-training treats pseudo labels produced by the model trained on labeled samples as ground truths of unlabeled samples. Therefore, the core of self-training is acquiring sufficient and accurate pseudo labels for the unlabeled samples. Conventionally, the pseudo labels are selected from the predictions with confidence higher than a certain threshold. In early works [15], [16], such threshold is usually set empirically. Moreover, it is inefficient because there are still many correctly predicted pixels in the pseudo-label images filtered out by this image-wise filtering. Later, to improve the filtering efficiency, some pixel-wise filtering methods [18], [19] are proposed. To acquire more accurate pseudo labels, stricter filters are set, like higher pixel-wise confidence. As the confidence increases, the utilization of unlabeled data will inevitably decrease, like Fig. 2. At the same time, there are a lot of semantic information is also lost during filtering. This limits further improvements in model performance. Therefore, how to balance the quality and quantity of pseudo-labels is a vexing problem.

Based on our analysis of consistency regularization and self-training, we propose a semi-supervised semantic segmentation method, namely HARC, for remote sensing images. It has two key parts, a hierarchical augmentation (HA) and the region-aware contrastive (RC) learning [28].

The HA consists of three paths to simulate three types of situations in remote sensing images. The first path uses only WA to simulate the first type of situation affected by spatial variations. The second path uses both WA and SA to simulate the second type of situation affected by spatial variations and uniform spectral variations. The third path uses a CutMix [29] between two the second type of situation to the third type of situation affected by spatial variations and uneven spectral variations. In our method, the first path is utilized for training on the labeled samples and generating pseudo labels. The second and third paths are utilized for training on the unlabeled samples.

How to balance the quality and quantity of pseudo-labels is a vexing problem. In terms of solving this problem, we did not do a lot of experiments to find its optimal confidence. We retain all pseudo labels to fully exploit unlabeled data. This would cause the model to be affected by numerous noisy labels during that unlabeled data training phase. Therefore, making the model learn the true feature distribution of unlabeled data is very critical. To solve this problem, we design a training module, namely RC. First, RC is used on the labeled data to construct a class-wise feature memory, enabling the learning of a genuine class-wise feature distribution. Then, RC is used on the unlabeled data, and it enables features extracted from unlabeled data to have the same distribution as features extracted from labeled data. It can make the feature extracted have the same class-wise feature distribution as the labeled data. Specifically, each picture is denoted by n regional centers based on the prediction and the feature of the picture. The n equals the number of categories in the dataset. Then, RC learning [28] is used to allow regional centers of the same category to be clustered together. In this way, a class-wise feature memory is found and updated in the labeled data training. Benefiting from accurate label guidance, the class-wise feature memory of labeled data has a more accurate distribution than the class-wise feature memory of unlabeled data. Therefore, the class-wise feature memory could be a good supervision for the unlabeled data to alleviate the impact of false pseudo labels. Compared with cross-entropy loss, it is a soft supervision so that it can be more robust under the interference of noisy labels.

Here are the main contributions we have made.

1) A HA that simulates three types of situations in remote sensing images is proposed. It is proven to be good for enhancing performance in semi-supervised semantic segmentation.
2) RC learning is introduced to fully and better utilize the unlabeled data. A class-wise feature memory, i.e., the center memory is constructed to reduces the influence of noisy pseudo labels on model performance.

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3) A new semi-supervised semantic segmentation method is proposed, named HA and RCl earning. Sufficient experiments prove its superiority over state-of-the-art (SOTA) methods.

II. Related Work

A. Consistency Regularization

Consistency regularization [30], [31] first applies various perturbations on an image to obtain a training sample pair, and then forces the model to get consistent predictions on the pair. In this way, consistency regularization can help the model learn the consistency distribution of the dataset. In these methods, the perturbations are the key. In some early works, the perturbations are some common data augmentations, like rotate and resize. These common data augmentations, usually called WA in semi-supervised learning, are very limited. Then, French et al. [7] did a large of experiments and proposed that semi-supervised semantic segmentation needs strong, high-dimensional perturbations, usually called SA in semi-supervised learning. Zhao et al. [9] believed that not all unlabeled samples are suitable for strong perturbations. They did some experiments and found that applying strong perturbations to simple samples would help enhance the model's robustness; however, applying strong perturbations to difficult samples will increase the difficulty of model training and make it difficult for the model to converge. In this work [9], they proposed a method to determine the difficulty of unlabeled samples and decide whether to use strong perturbations on the samples based on the difficulty. To ensure the model prioritizes difficult samples, Xin et al. [13] used contrastive learning to assign higher weights to them. To obtain more different augmented images, Yang et al. [14] enhanced unlabeled images by randomly copying and pasting class regions from labeled images. In another work [8], they proposed a method that applied Cutmix between labeled samples and difficult unlabeled samples to decrease the impact of difficult unlabeled samples. Since then, these works added image-wise perturbations to the input. Yang et al. [32] proposed an auxiliary feature-wise perturbation as a supplement. To produce new training data in remote sensing, Lu et al. [10] proposed a sparse dual-view cross-sample image generation method. For remote sensing data, WA and SA could be, respectively, seen as the spatial variation and the uneven spectral variation. However, many scenarios are often subject to uneven spectral variations. Therefore, it is of great significance to design a HA to allow the model to adapt to remote sensing data in various situations.

B. Self-Training

Self-training employs the model trained on the labeled data to produce pseudo labels, and then these pseudo labels are used in supervising the unlabeled data. Accuracy and quantity of pseudo labels are the keys to determining the model's performance. In semi-supervised semantic segmentation, the labeled data is insufficient to train a satisfactory model, thus leading to noisy pseudo labels. Therefore, most of the self-training methods focus on how to filter noise pseudo labels. To improve the accuracy of pseudo labels, early works [15], [16] often set a manual image-wise confidence to filter noisy pseudo labels. It relies too much on the designer's experience and is not efficient, because there are some correct pixel-wise pseudo labels in the low-confidence image-wise pseudo labels. To reduce the loss of correct pseudo labels, many pixel-wise filters are proposed [11], [18], [19], [20], [21]. Li et al. [18] used generative adversarial network [33] to train a pixel-level discriminator to filter to get a more accurate pseudo-label. Lu et al. [11] proposed a linear sampling strategy for assigning thresholds to different classes to filter the noise. In addition, based on a pixel-level filter, Li et al. [26] also utilized a boundary filter to obtain more accurate pseudo labels. Huang et al. [27] proposed a category-adaptive threshold selection method to alleviate the class imbalance issue during the pseudo-label filtering process. Because pseudo labels' quantity will be affected in the process of improving pseudo labels' accuracy, these methods require a large number of experiments to find the balance between pseudo labels' quantity and pseudo labels' accuracy. This is time-consuming and empirical. To effectively utilize all pseudo labels, it is imperative to identify an approach that is resistant to noise pseudo-label interference.

C. Contrastive Learning

Contrastive learning is a self-supervised learning method that allows the model to learn robust features by making positive samples closer and pushing negative samples further away. In the early years, contrastive learning achieved many breakthroughs [34], [35], [36], [37], [38]. Contrastive learning has a high dependence on numerous negative pairs. However, numerous negative pairs consume a lot of computer memory. To reduce the consumption of computer memory, He et al. [36] proposed momentum contrast which used the momentum update between two encoders in the contrastive learning paradigm. Early works [34], [35] require a specialized architecture that leads the contrastive learning complex. To simplify the use of contrastive learning, Chen et al. [37] proposed to use cosine similarity to measure positive and negative pairs. These methods all used image-level contrastive learning. For semantic segmentation, there are many pixel-level contrastive learning methods [39], [40], [41] proposed. The number of samples for pixel-level contrastive learning is tens or even hundreds of times that of image-level contrastive learning. So, it is time-consuming and memory-consuming. Hu et al. [28] proposed a RC learning method using a region center to represent pixel-level features to make the memory bank lightweight. In RC learning, first, they classify each pixel feature based on the prediction. Then, the weighted average of the same type of pixel features on the same picture is used as a region center. Finally, they use contrastive learning on region centers across the entire dataset to allow the model to learn the deep class-wise feature distribution of the dataset. In this way, they improve the performance of the model in supervised semantic segmentation. However, this method cannot be without accurate labels, because these labels can help it obtain accurate predictions to reduce the error rate in

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feature classification. Therefore, in our method, we just save and update the region centers of labeled data. For unlabeled data, their region centers are judged by the region centers of labeled data. In this way, we believe that the region centers of labeled data are a type of accurate and robust class-wise feature distribution of the dataset. Using them to supervise the learning on the unlabeled data can alleviate the confirmation bias [42] which may make the network outfit the noisy label.

III. Proposed Method

An overview of HARC is depicted in Fig. 3. HARC utilizes a teacher–student framework with exponential moving average (EMA). It consists of two main components: supervised learning and unsupervised learning. A labeled dataset $D_l = \{x_l, y_l\}^{N_l}$ and an unlabeled dataset $D_u = \{x_u\}^{N_u}$ are, respectively, used in training on these two parts, where N_l and N_u denote the quantities of D_l and D_u, respectively. x_l and x_u denote the labeled and unlabeled images, respectively, and y_l denotes the labels corresponding to x_l. Two modules, the HA module and the RC module, play crucial roles in this framework. HA enhances N_u by enabling the network to fully learn the impact of various variations on the image. RC enhances the network's performance in supervised learning and mitigates the impact of noisy pseudo-labels in unsupervised learning. Specifically, in supervised learning, HA first applies WA before the network. Then, RC uses features f_l and predictions p_l to obtain region centers $\{R_i\}_l$. After that, a center memory M has been instructed. Finally, $\{R_i\}_l$ and p_l are used in computing contrastive loss and pixel-level cross-entropy loss with M and y_l. Behind each epoch of supervised learning, M is updated. In unsupervised learning, first, two versions of x_u are obtained by HA. These two versions both have WA and SA operations applied. The second version has been also subjected to Cutmix with the first. In this way, N_u has been improved. Then, region centers and predictions of these two versions, are obtained. Finally, region centers and predictions are used in computing RC loss and pixel-level cross-entropy loss with M and pseudo labels. Because pseudo labels are difficult to be as accurate as labels, M is not updated in unsupervised learning.

A. HA Module

Image Augmentation is an important way to enhance the model's robustness. Recently, in semi-supervised semantic segmentation, weak-to-strong consistency regularization methods have achieved great success [8], [9], [32]. In remote sensing, WA can simulate the impact of spatial variations. And SA can simulate the impact of uniform spectral variations. In fact, there are some uneven spectral variations that need to be simulated. Therefore, to allow the model to fully learn different scenarios of remote sensing images, we propose the HA module.

As shown in Fig. 3, HA has three paths and is composed of WA, SA, and Cutmix. HA is applied before the student network. In this module, the labeled and unlabeled images are transformed in different ways to produce three different augmented versions

$$\hat{x}_l, \hat{x}_u, \hat{x}'_u = H(x_l, x_u) \tag{1}$$

where H is hierarchical augmentation.

Specifically, the first path H_1 is designed for x_l. It can be denoted by the following equation:

$$\hat{x}_l = H_1(x_l) = A_w(x_l) \tag{2}$$

where A_w is weak augmentation.

The other paths are designed for x_u. They can improve the utilization rate of x_u. These paths fully simulate the impact of spectral variations. Following equation is the second path H_2 that simulates the impact of uniform spectral variations:

$$\hat{x}_u = H_2(x_u) = A_w(A_s(x_u)) \tag{3}$$

where A_s is strong augmentation.

Following equation is the third path H_3 that simulates the impact of uneven spectral variations:

$$\begin{aligned}\hat{x}'_u &= H_3(x_u) \\ &= A_c\left(H_2(x_u), A'_w\left(A'_s(x_u)\right)\right)\end{aligned} \tag{4}$$

where A'_w and A'_s are weak augmentation and strong augmentation in H_3, and A_c is Cutmix.

Notably, we only use unlabeled data to simulate the impact of spectral variations. On labeled data training, there are some refined manual labels to supervise training. The accuracy of manual labels is much higher than that of pseudo labels. This means that the prediction requirements for labeled data are much higher than those for unlabeled data under the same augmentations. Therefore, if complex augmentations are added during the training process of labeled data, it will greatly increase the difficulty of model training. In semi-supervised learning, to maintain a balance in difficulty between labeled and unlabeled data, complex augmentations such as SA need to be applied to the unlabeled data.

B. RC Module

First, we review the RC learning [28]. It is a successful leap to advance contrastive learning from the image level to the region level or pixel level in supervised semantic segmentation. It uses a region center to represent the features of all pixels of the same category in an image. In an individual image, the region center of class i is the average of the features of each pixel assigned to class i

$$R_i = \frac{\sum_{(a,b)} F_{(a,b)} \mathbf{1}[L(a,b) = i]}{\sum_{(a,b)} \mathbf{1}[L(a,b) = i]} \tag{5}$$

where the $F_{(a,b)}$ denotes feature of pixel (a, b) and the L denotes the prediction inferred by the student network.

After performing the above operations on the entire dataset, the RC loss for each region center of class i is

$$L_i^{\text{NCE}} = \frac{1}{|M_i|} \sum_{k_+ \in M_i} -\log \frac{\exp(q \cdot k_+/\tau)}{\exp(q \cdot k_+/\tau) + \sum_{k_-} \exp(q \cdot k_-/\tau)} \tag{6}$$

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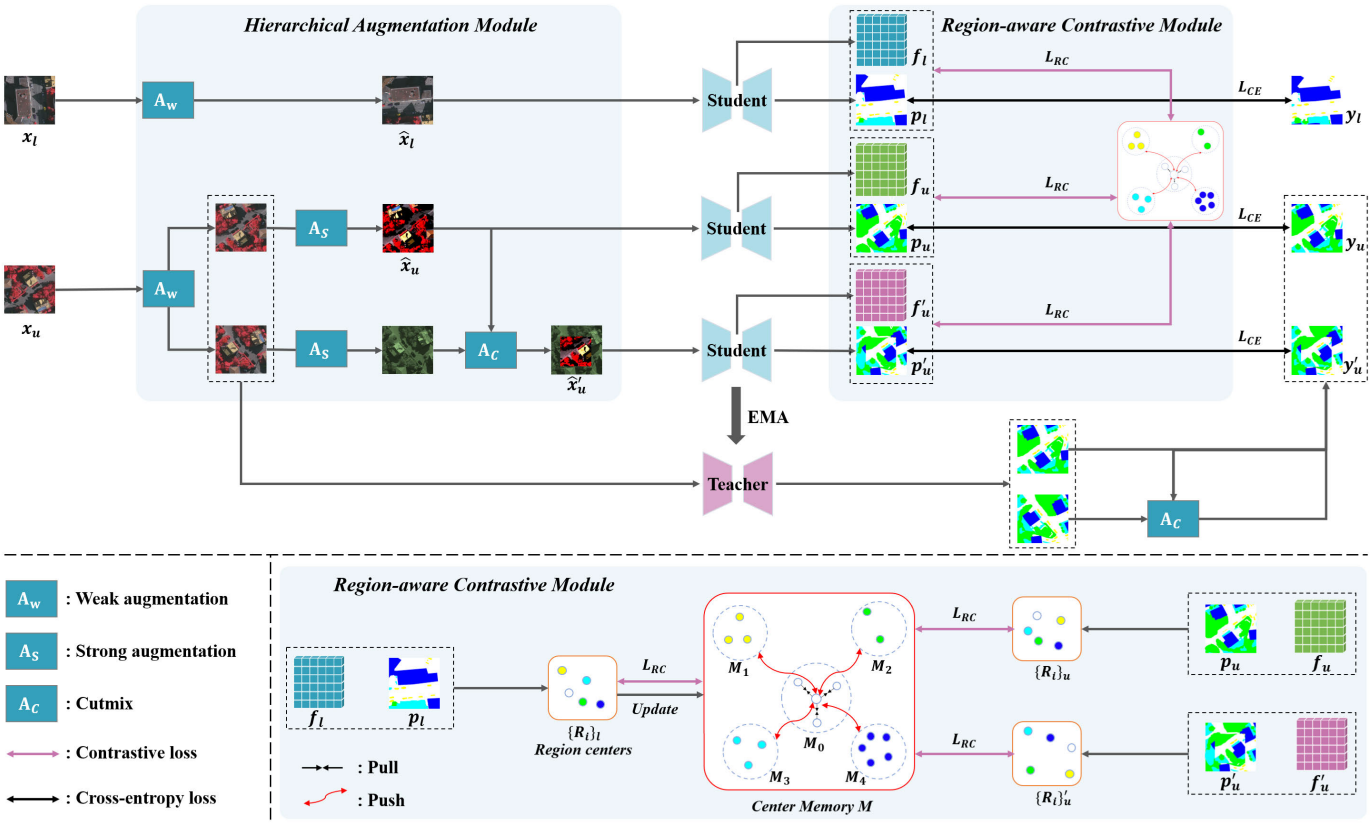


Fig. 3. Overview of HARC. x_l and x_u are labeled images and unlabeled images. y_l is the label of x_l. y_u and y'_u are the pseudo labels. f_l, f_u, and f'_u are the features extracted by the segment network. $\hat{x}_l$ is the image affected by spatial variations. $\hat{x}_u$ is the image affected by spatial variations and uniform spectral variations. $\hat{x}'_u$ is the image affected by spatial variations and uneven spectral variations. EMA is the exponential moving average.

TABLE I
PERFORMANCE COMPARISON STUDY ON THE POTSDAM DATASET

Labeled Data	Method	F1					Mean F1	Mean IoU
		Surface	Building	Low-Veg	Tree	Car		
1/8	baseline	87.26	91.29	80.67	80.55	73.03	82.56	70.82
	Hung et al. [43]	-	-	-	-	-	-	-
	S4-GAN [19]	88.64	92.01	81.36	81.93	84.09	85.61	75.62
	CCT [22]	87.31	91.12	80.32	81.62	79.36	83.95	72.60
	Li et al. [18]	89.18	92.59	82.32	83.39	86.63	86.82	76.92
	WSCL [10]	87.89	91.68	81.09	82.75	89.55	86.59	76.57
	Unimatch [32]	88.61	92.60	**82.45**	**83.73**	90.34	87.55	78.06
	HARC	**89.27**	**93.05**	81.74	83.55	**90.71**	**87.66**	**78.30**
1/4	baseline	88.82	92.72	82.49	82.09	79.31	85.09	74.37
	Hung et al. [43]	-	-	-	-	-	-	-
	S4-GAN [19]	88.81	92.92	81.20	80.78	85.09	85.76	76.62
	CCT [22]	88.33	93.00	81.26	83.04	82.34	85.59	75.09
	Li et al. [18]	**89.67**	**93.83**	82.75	82.22	86.92	87.08	77.39
	WSCL [10]	88.58	92.43	82.57	83.76	89.49	87.37	77.75
	Unimatch [32]	89.43	92.83	**83.49**	84.97	90.40	88.22	79.10
	HARC	89.41	93.23	82.95	**85.26**	**91.16**	**88.40**	**79.42**

where the M_i denotes the all-region centers that belongs to class i in the dataset and k_- is another class in the M_i.

In our method, RC learning is applied to obtain reliable class-wise feature distribution which is effective in improving the performance of the segment network. Therefore, using RC loss and pixel-level cross-entropy loss jointly in unsupervised learning can alleviate the impact of noisy pseudo labels. Specifically, we first use RC learning in supervised learning. Then, all region centers of labeled data form a center memory M

$$M = \{M_i | i = 0, 1, 2, \ldots, n_c\} \tag{7}$$

$$M_i = \left\{R_i^x \middle| x \in D_l\right\} \tag{8}$$

where n_c is the number of categories.

After that, we would computer region centers of all unlabeled data. Finally, the region contrastive loss is computed

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between region centers of all unlabeled data and M. M is just updated in supervised learning.

C. Loss Function

In HARC, pixel-level cross-entropy loss and RC loss are applied. So, our total loss function is as formula (3)

$$L = L_{\mathrm{CE}} + \lambda L_{\mathrm{RC}} \tag{9}$$

where L_{CE} and L_{RC} are pixel-level cross-entropy loss and RC loss, respectively.

1) Pixel-Level Cross-Entropy Loss: The pixel-level cross-entropy loss is the commonest loss function in semantic segmentation. It considers whether the classification is right. In our method, L_{CE} consists of supervised-learning section $L_{\mathrm{CE}}^{\mathrm{sup}}$ and unsupervised-learning section $L_{\mathrm{CE}}^{\mathrm{unsup}}$

$$L_{\mathrm{CE}}^{\mathrm{sup}} = -\frac{1}{|N_l|}\sum_{x_l \in D_l} y_l \log(S(\hat{x}_l)) \tag{10}$$

where S is the student network

$$L_{\mathrm{CE}}^{\mathrm{unsup}} = -\frac{1}{2|N_u|}\sum_{x_u \in D_u} \left(T(\hat{x}_u)\log(S(\hat{x}_u)) + T\left(\hat{x}'_u\right)\log\left(S\left(\hat{x}'_u\right)\right)\right) \tag{11}$$

where T is the teacher network

$$L_{\mathrm{CE}} = L_{\mathrm{CE}}^{\mathrm{sup}} + L_{\mathrm{CE}}^{\mathrm{unsup}}. \tag{12}$$

2) RC Loss: The RC loss L_{RC} also consists of two sections, $L_{\mathrm{CE}}^{\mathrm{sup}}$ and $L_{\mathrm{CE}}^{\mathrm{unsup}}$, as same as L_{CE}

$$L_{\mathrm{RC}}^{\mathrm{sup}} = \frac{1}{|N_l|}\sum_{x_l \in D_l}\sum_{i}^{n_c} L_i^{\mathrm{NCE}} \tag{13}$$

$$L_{\mathrm{RC}}^{\mathrm{unsup}} = \frac{1}{2|N_u|}\left(\sum_{\hat{x}_u}\sum_{i}^{n_c} L_i^{\mathrm{NCE}} + \sum_{\hat{x}'_u}\sum_{i}^{n_c} L_i^{\mathrm{NCE}}\right) \tag{14}$$

$$L_{\mathrm{RC}} = L_{\mathrm{RC}}^{\mathrm{sup}} + L_{\mathrm{RC}}^{\mathrm{unsup}}. \tag{15}$$

IV. Experiments

In the experiment, several SOTA methods are compared with HARC, including three consistency regularization methods [10], [22], [32] and three self-training methods [18], [19], [43]. We additionally perform numerous ablation experiments to illustrate how both the HA module and the RC module contribute to our method's efficacy.

A. Datasets

In these experiments, HARC's performance is assessed on three datasets: two from remote sensing (ISPRS), namely Potsdam[1] and Vaihingen,[2] and a Land-cover dataset called LoveDA [44].

Potsdam: Potsdam is one of the most commonly used datasets in the semantic segmentation of RS images. The dataset was collected in the urban area of Potsdam, Germany. To promote the progress of semantic segmentation of high-resolution RS images, the spatial resolution of each image can reach 5 cm/pixel. The researchers carefully annotated these images and divided them into six categories, namely impervious surfaces, building, low vegetation, tree, car, and background. The entire dataset consists of 38 images with a width of 6000 × 6000 pixels. In the experiment, the first 24 pictures are used for training and the last 14 pictures are used for testing.

Vaihingen: Vaihingen is also a set of aerial RS image datasets, which has the same annotation category as the former. It consists of 33 images, ranging in size from 1996 × 1995 pixels to 3816 × 2550 pixels. Sixteen of them are used for training and 17 are used for testing.

LoveDA: LoveDA is a land-cover dataset constructed by the Wuhan University. It contains 5987 images with 0.3 m high-resolution and 166 768 annotated semantic objects from three different cities: Nanjing, Changzhou, and Wuhan. The image sizes are all 1024 × 1024 pixels. It contains both urban and rural areas, which brings considerable challenges: 1) objects of varying scales; 2) intricate background patterns; and 3) inconsistent sample distribution. The training set comprises 1156 urban images and 1366 rural images, while the test set includes 677 urban images and 992 rural images.

In our experiments, all images are cropped to 512 × 512 pixels. For the Potsdam and Vaihingen datasets, the overlap when cropping is 128 pixels, while the overlap of LoveDA is 0. Taking the Potsdam dataset as an example, the cropped training set contains a total of 6144 images. If the first 1/8 of the images are used as the labeled part, then the labeled part contains 768 images and the unlabeled part has 5376 images. To facilitate the adjustment of parameters, we simply copy the image with the label part seven times, so that it has the same number of images as the unlabeled part.

B. Experimental Setup

Considering the experiment and our experimental conditions, FCN [3] with resnet-50 [2] pretrained by ImageNet, hereafter referred to as FCN-resnet50, is set as the teacher network and the student network. We used two Nvidia RTX2080Ti to complete the experiments on Potsdam and Vaihingen and used an Nvidia RTX3090 GPU to complete the experiments on LoveDA. Each batch contains four labeled images and four unlabeled images. We set the optimizer to be the same as [18].

C. Evaluation Metric

We use the $F1$ and Mean IoU to evaluate the performance of the methods

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{16}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{17}$$

$$F1 = 2 \times \frac{P \times R}{P + R} \tag{18}$$

[1]http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-potsdam.html

[2]http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

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TABLE II
PERFORMANCE COMPARISON STUDY ON THE VAIHINGEN DATASET

Labeled Data	Method	F1					Mean F1	Mean IoU
		Surface	Building	Low-Veg	Tree	Car		
1/8	baseline	80.86	83.71	71.06	82.26	44.77	72.53	58.74
	Hung et al. [43]	82.85	87.17	**74.95**	**84.14**	0.02	65.83	56.37
	S4-GAN [19]	83.09	84.97	73.31	83.06	60.28	77.05	63.40
	CCT [22]	81.30	84.66	69.64	81.72	40.71	71.61	57.99
	Li et al. [18]	83.03	86.33	73.57	83.72	63.24	77.98	64.68
	WSCL [10]	83.57	87.38	73.83	81.90	56.80	76.70	63.38
	Unimatch [32]	86.57	89.93	67.39	78.70	67.18	77.95	64.86
	HARC	**86.67**	**90.83**	74.53	83.58	**71.31**	**81.38**	**69.26**
1/4	baseline	83.80	87.05	74.05	81.82	58.21	76.99	63.66
	Hung et al. [43]	84.78	88.50	74.91	82.91	0.07	66.23	56.74
	S4-GAN [19]	85.44	88.78	75.61	82.91	66.86	79.92	67.25
	CCT [22]	85.16	89.84	76.75	82.25	54.14	77.63	64.99
	Li et al. [18]	85.67	88.96	76.11	83.48	68.69	80.58	68.29
	WSCL [10]	85.61	89.06	75.25	83.24	66.12	79.86	67.22
	Unimatch [32]	86.78	90.62	75.75	83.48	72.05	81.74	69.69
	HARC	**88.08**	**91.54**	**78.20**	**84.55**	**75.82**	**83.64**	**72.32**

TABLE III
PERFORMANCE COMPARISON STUDY ON THE LOVEDA DATASET

Labeled Data	Method	F1							Mean F1	Mean IoU
		Background	Building	Road	Water	Barren	Forest	Agriculture		
1/8	baseline	65.71	66.44	62.17	76.14	39.82	54.03	59.05	60.48	44.15
	Li et al. [18]	66.31	69.53	64.64	75.50	39.57	**57.44**	64.10	62.44	46.20
	WSCL [10]	62.92	67.20	61.90	59.51	11.59	29.10	50.52	48.96	34.38
	Unimatch [32]	**66.54**	72.95	64.64	73.89	40.16	51.41	**65.51**	62.16	46.01
	HARC	66.31	**73.41**	**67.22**	**77.58**	**40.36**	55.56	63.62	**63.44**	**47.43**
1/4	baseline	67.30	70.07	65.08	**79.11**	38.26	54.56	62.42	62.40	46.41
	Li et al. [18]	67.41	69.58	63.43	78.61	35.96	55.88	64.22	62.16	46.20
	WSCL [10]	63.28	64.45	65.75	67.97	20.75	32.91	60.18	53.61	38.37
	Unimatch [32]	65.73	75.23	65.63	72.39	39.48	52.07	**64.57**	62.16	46.04
	HARC	**68.49**	**75.57**	**69.83**	77.70	**41.93**	**59.70**	64.36	**65.37**	**49.50**

TABLE IV
PERFORMANCE ABLATION STUDY ON THE VAIHINGEN DATASET

Labeled Data	Method	F1					Mean F1	Mean IoU
		Surface	Building	Low-Veg	Tree	Car		
1/8	baseline	80.86	83.71	71.06	82.26	44.77	72.53	58.74
	WSA	86.65	90.23	73.67	82.30	71.12	80.79	68.41
	WSA+RC	**86.95**	90.02	**74.78**	**84.10**	69.96	81.16	68.96
	HA	86.67	90.61	73.92	82.77	70.55	80.90	68.61
	HARC	86.67	**90.83**	74.53	83.58	**71.31**	**81.38**	**69.26**
1/4	baseline	83.80	87.05	74.05	81.82	58.21	76.99	63.66
	WSA	86.45	89.87	74.63	83.12	72.62	81.34	69.07
	WSA+RC	87.99	91.38	77.59	84.42	75.44	83.36	71.93
	HA	87.07	90.22	77.34	**84.66**	73.45	82.54	70.76
	HARC	**88.08**	**91.54**	**78.20**	84.55	**75.82**	**83.64**	**72.32**

$$\text{Mean IoU} = \frac{1}{N_c} \times \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{19}$$

where N_c is the number of categories, TP and TN are the numbers of true positive and true negative, and FP and FN are the numbers of false positive and false negative.

D. Compare With Baseline

Tables I–III, respectively, show the experiment results on the Potsdam, Vaihingen, and LoveDA test sets. On these three datasets, HARC shows SOTA performance.

Compared with the baseline, HARC has a great improvement on these three test sets. Using 1/8 of the labeled data, HARC improves the Mean $F1$ over the baseline by 5.1%, 8.85%, and 2.96% on the Potsdam, Vaihingen, and LoveDA datasets, respectively, and the mean IoU by 7.48%, 10.52%, and 3.28% on these datasets, respectively. Using 1/4 of the labeled data, HARC improves the mean $F1$ over the baseline by 4.31%, 6.65%, and 2.97% on the Potsdam, Vaihingen, and LoveDA datasets, respectively, and the mean IoU by 5.05%, 8.66%, and 3.09% on these datasets,

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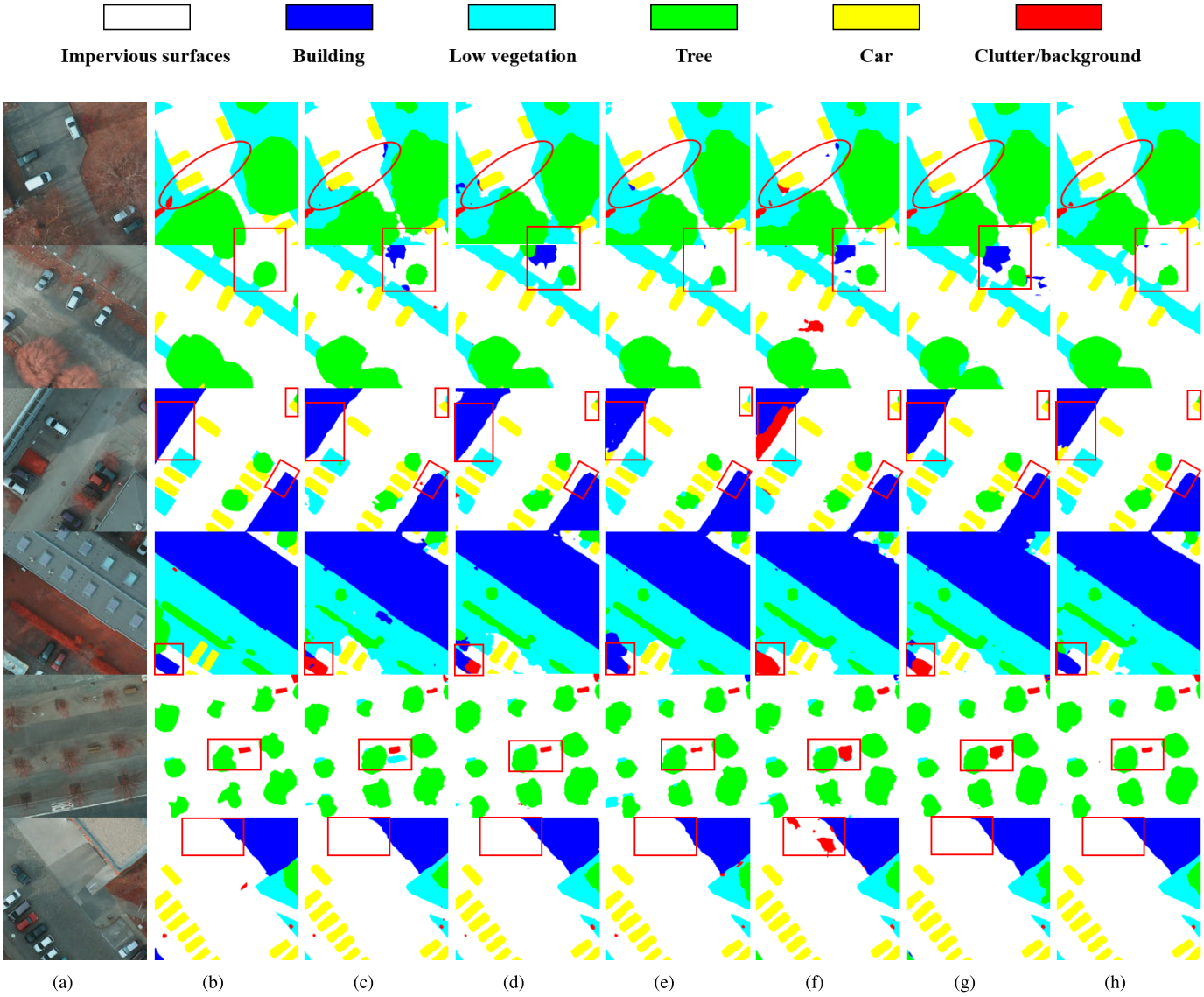


Fig. 4. Comparison on Potsdam test set (1/8 labeled data and 1/4 labeled data). (a) Input image. (b) Ground truth. (c) WSCL trained on 1/8 labeled data. (d) Unimatch trained on 1/8 labeled data. (e) HARC trained on 1/8 labeled data. (f) WSCL trained on 1/4 labeled data. (g) Unimatch trained on 1/4 labeled data. (h) HARC trained on 1/4 labeled data.

respectively. On the Potsdam and Vaihingen test sets, the segmentation effect of the car category has been improved the most. It can be seen from the baseline evaluation that the car category is the most difficult to segment. And among these two datasets, the car category has the smallest number of training samples. It shows that HARC has a better performance in the small and difficult category. Among these three datasets, the Vaihingen dataset has the fewest training samples, while HARC has the greatest improvement on the Vaihingen. This shows that HARC can better cope with the challenge of a small number of labeled samples.

E. Compare With Semi-Supervised Works

Figs. 4 and 5 show comparative segmentation results on the Potsdam and Vaihingen datasets, respectively. Compared with methods based on consistency regularization, like CCT [22], WSCL [10], and Unimatch [32]. Using 1/8 of the labeled data, HARC improves the Mean $F1$ over other methods by 0.11%–3.71%, 3.43%–9.77%, and 1.28%–14.48% on the Potsdam, Vaihingen, and LoveDA datasets, respectively. It also enhances the Mean IoU by 0.24%–5.7%, 4.4%–11.27%, and 1.42%–13.05% on these datasets, respectively. Using 1/4 of the labeled data, HARC improves the Mean $F1$ over other methods by 0.18%–2.81%, 1.9%–6.01%, and 3.21%–11.76% on the Potsdam, Vaihingen, and LoveDA datasets, respectively. It also enhances the Mean IoU by 0.32%–4.33%, 2.63%–7.33%, and 4.06%–11.13% on these datasets, respectively.

Compared with methods based on self-training, like Hung et al. [43], s4-GAN [19], and Li et al. [18]. Using 1/8 of the labeled data, HARC improves the mean $F1$ over other methods by 0.84%–2.05%, 3.4%–15.55%, and 1.00% on the Potsdam, Vaihingen, and LoveDA datasets, respectively. It also enhances the mean IoU by 1.38%–2.68%, 4.58%–12.89%, and 1.23% on these datasets, respectively. Using 1/4 of the labeled data, HARC improves the mean $F1$ over other methods by 1.32%–2.64%, 3.06%–17.41%, and 3.21% on the Potsdam, Vaihingen, and LoveDA datasets, respectively. It also enhances

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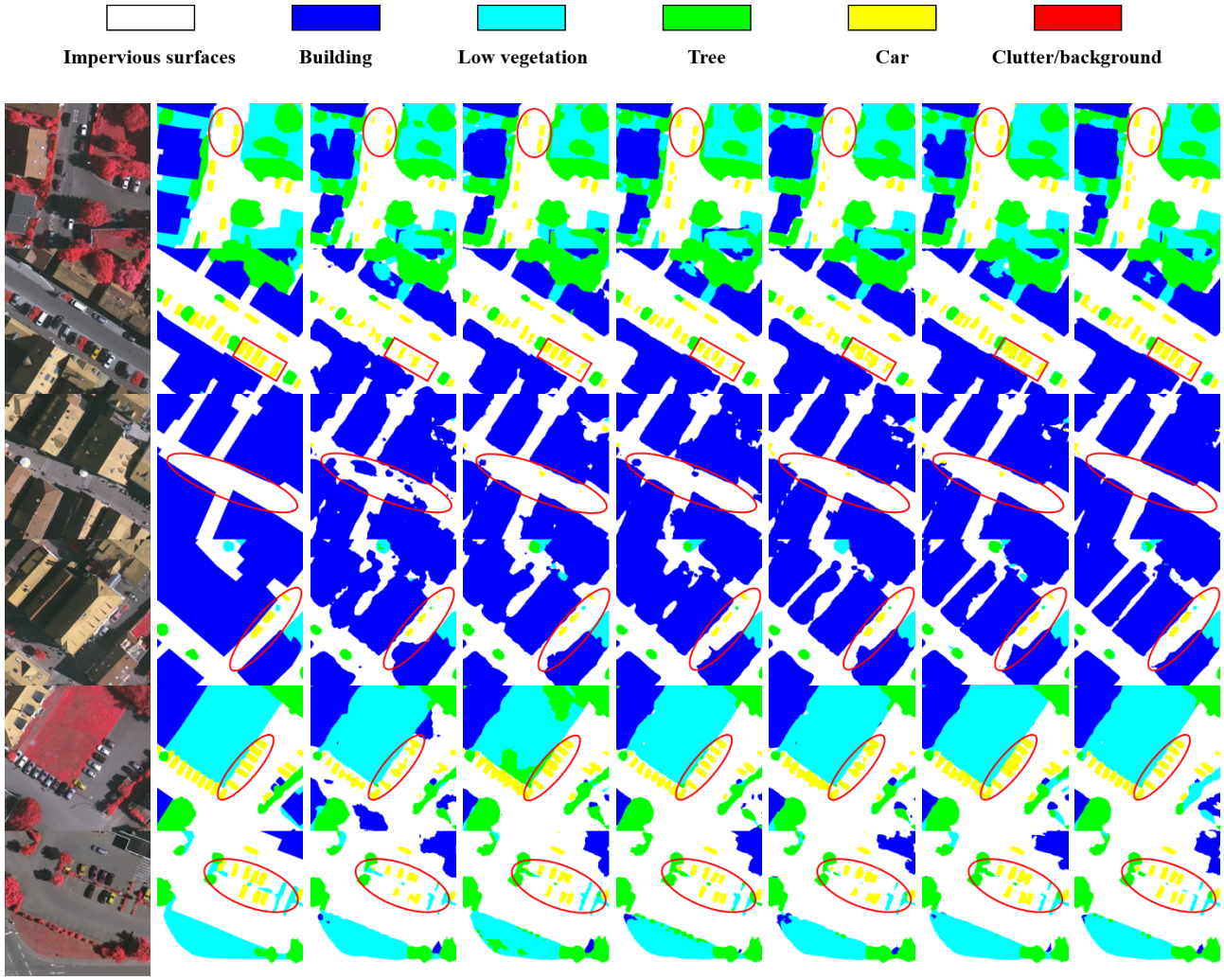


Fig. 5. Comparison on Vaihingen test set (1/8 labeled data and 1/4 labeled data). (a) Input image. (b) Ground truth. (c) WSCL trained on 1/8 labeled data. (d) Unimatch trained on 1/8 labeled data. (e) HARC trained on 1/8 labeled data. (f) WSCL trained on 1/4 labeled data. (g) Unimatch trained on 1/4 labeled data. (h) HARC trained on 1/4 labeled data.

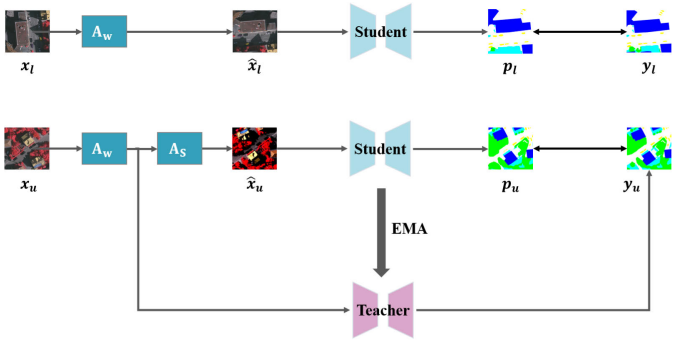


Fig. 6. Overview of WSA. x_l and x_u are labeled images and unlabeled images. p_l and p_u are the prediction of x_l and x_u. y_l are labels of x_l. y_u are pseudo labels.

the mean IoU by 2.03%–2.80%, 4.03%–15.58%, and 3.30% on these datasets, respectively.

F. Ablation Study

We have finished the ablation experiments on the Vaihingen dataset. The ablation experiments explored the effects of HA and RC, respectively. Table IV shows the results. Fig. 6 shows the overview of weak-to-SA (WSA). WSA + RC is WSA with the RC module. HA is HARC without the RC module.

The comparison between HA and WSA, as well as between HARC and WSA + RC, demonstrates the superior performance of HA. Under the condition of using 1/8 of the labeled data, the comparison between HA and WSA shows an improvement in the Mean $F1$ by 0.21% and in the mean IoU by 0.20%. Similarly, the comparison between HARC and WSA + RC shows an improvement in the Mean $F1$ by 0.22% and in the Mean IoU by 0.30%. Under the condition of using 1/4 of the labeled data, the comparison between HA and WSA shows an improvement in the Mean $F1$ by 1.2% and in the Mean IoU by 1.69%. Similarly, the comparison between HARC and WSA + RC shows an improvement in the Mean $F1$ by 0.28% and in the Mean IoU by 0.39%.

The comparison between WSA + RC and WSA, as well as between HARC and HA, demonstrates the superior performance of RC. Under the condition of using 1/8 of the labeled data, the comparison between WSA + RC and WSA shows an improvement in the Mean $F1$ by 0.27% and in the Mean IoU by 0.55%. Similarly, the comparison between HARC and

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HA shows an improvement in the Mean $F1$ by 0.48% and in the Mean IoU by 0.65%. Under the condition of using 1/4 of the labeled data, the comparison between WSA + RC and WSA shows an improvement in the Mean $F1$ by 2.02% and in the Mean IoU by 2.86%. Similarly, the comparison between HARC and HA shows an improvement in the Mean $F1$ by 1.10% and in the Mean IoU by 1.56%. In comparison, we found that the RC module achieved a greater performance improvement on 1/4 of labeled data than that on 1/8 of labeled data. The reason may be that more labeled data allows the center memory to obtain more region centers, which benefits the region contrast process. These improvements show that the RC module has a good effect on semi-supervised semantic segmentation in RS. And the improvements of RC are more significant than HA.

V. Conclusion

In this work, we present a new semi-supervised semantic segmentation method for remote sensing images, namely HARC. It features two innovative modules: the HA module and the RC module. The HA module can simulate a broader range of situations compared to traditional weak-to-strong consistency regularization methods. This allows the segmentation network to learn from images impacted by uneven spectral variations. The RC module enables the segmentation network to fully capture the class-wise feature distribution of labeled data. This learned distribution can then supervise the training on unlabeled data, thereby alleviating the impact of noisy pseudo labels. We evaluated our method on three publicly available remote sensing datasets. The experimental results show that HARC surpasses many SOTA methods. Additionally, ablation studies demonstrate the effectiveness of both the HA module and the RC module.

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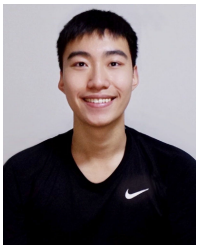
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